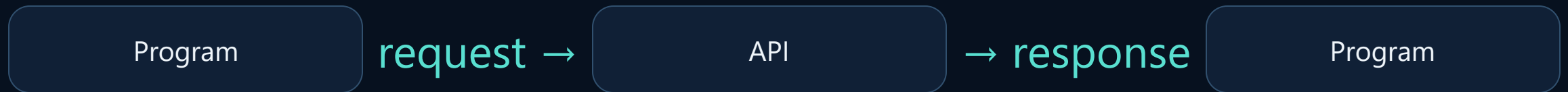


APIs, secrets, and failures.

Know what crosses the boundary. Protect access. React by evidence.



**TODAY'S
OUTCOME**

Inspect, protect, classify, act.

Inspect

Request and
response

Protect

Keys and private
data

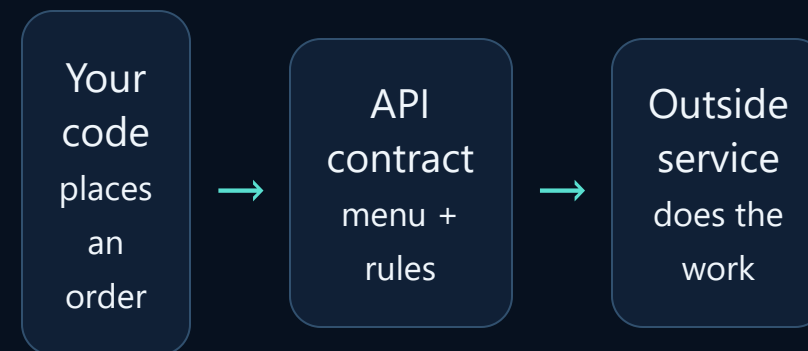
Classify

Error layer

Act

Fix, retry, or stop

A documented counter between programs.



TWO
HALVES

Request out. Response back.

Request

Destination · authentication ·
headers · body · timeout

Response

Status · content or error ·
headers · usage

HEADERS

Metadata about the exchange.

Authorization

Identifies and permits the caller.

Content-Type

Explains how the body is encoded.

Log the header name if useful. Never log its secret value.

API
KEY

Access identity connected to usage.

**Grants
access**

**Tracks
usage**

**May
create
cost**

Leak? Revoke or rotate it, replace it privately, and inspect usage.

**BEFORE
SEND**

Read the request without revealing authentication.

Instruction

Messages

Limit

**Data
approval**

**AFTER
RECEIVE**

Read the response as evidence.

Content

Provider

Model

Usage

Cost

OUTPUT
LIMIT

A small limit can cut the answer short.

```
max_tokens = 15

if
reply.output_tokens
>= max_tokens:
    print("answer
may be cut off")
```

A warning signal, not perfect stop-reason proof.

**STATUS
CODES**

The number narrows the next action.

2xx

Request
succeeded

4xx

Request,
access, or limit

5xx

Outside service
problem

**ERROR
DECISION**

Fix or retry?

Fix first

400 · 401 · 403

Bad request, authentication,
permission.

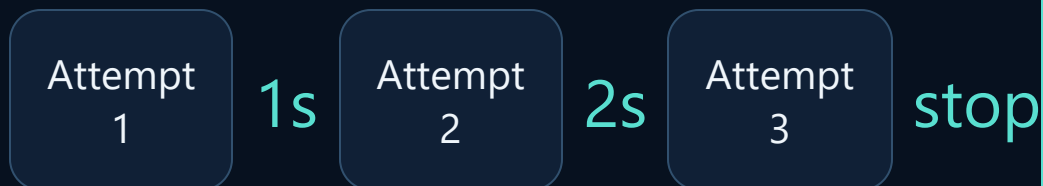
May retry

429 · 5xx · connection

Rate limit, service, temporary network.

BACKOFF

Wait longer instead of hammering.



Maximum
attempts +
timeout + small
random jitter.

IDEMPOTENCY

Would repeating create a duplicate action?

Usually safe

Read
weather ·
fetch order
· search
docs

Needs protection

Charge card ·
send email ·
create order

Use an idempotency
key or duplicate-
prevention record.

**ERROR
TAXONOMY**

Name the failing layer.

Input

Access

Input · authentication · permission ·
rate limit · network · provider · parsing
· programming

Outside

Code/data

FINISHED
HANDLER

Return useful advice without hiding failure.

```
try:  
    return ask(prompt)  
except Exception as  
error:  
    return  
friendly_message(error)
```

User-safe result +
restricted technical
evidence.

LAB
4.A

Inspect request and response parts.

```
system · messages · max_tokens
```



```
model · provider · content ·  
input/output tokens · cost
```

LAB
4.B

Translate and contain outside failure.

friendly_message

Error evidence → action

safe_ask

Try call →
catch →
return

**DETERMINISTIC
PROOF**

Force every branch locally.

400

401

Controlled fake errors. Real
translator. No key. No network.

403

429

500

**PRODUCTION
BOUNDARY**

Reliability needs limits and ownership.

Timeout

**Retry
budget**

Redaction

**Fallback
owner**

COMPLETION
GATE

Day 4 is complete when failure leads to the right action.

- ✓ Describe request and response fields.
- ✓ Keep credentials out of code and logs.
- ✓ Map controlled errors to useful actions.
- ✓ Retry only temporary, safe operations.
- ✓ Record timeout, attempt limit, and fallback.

Next: files and document ingestion.